

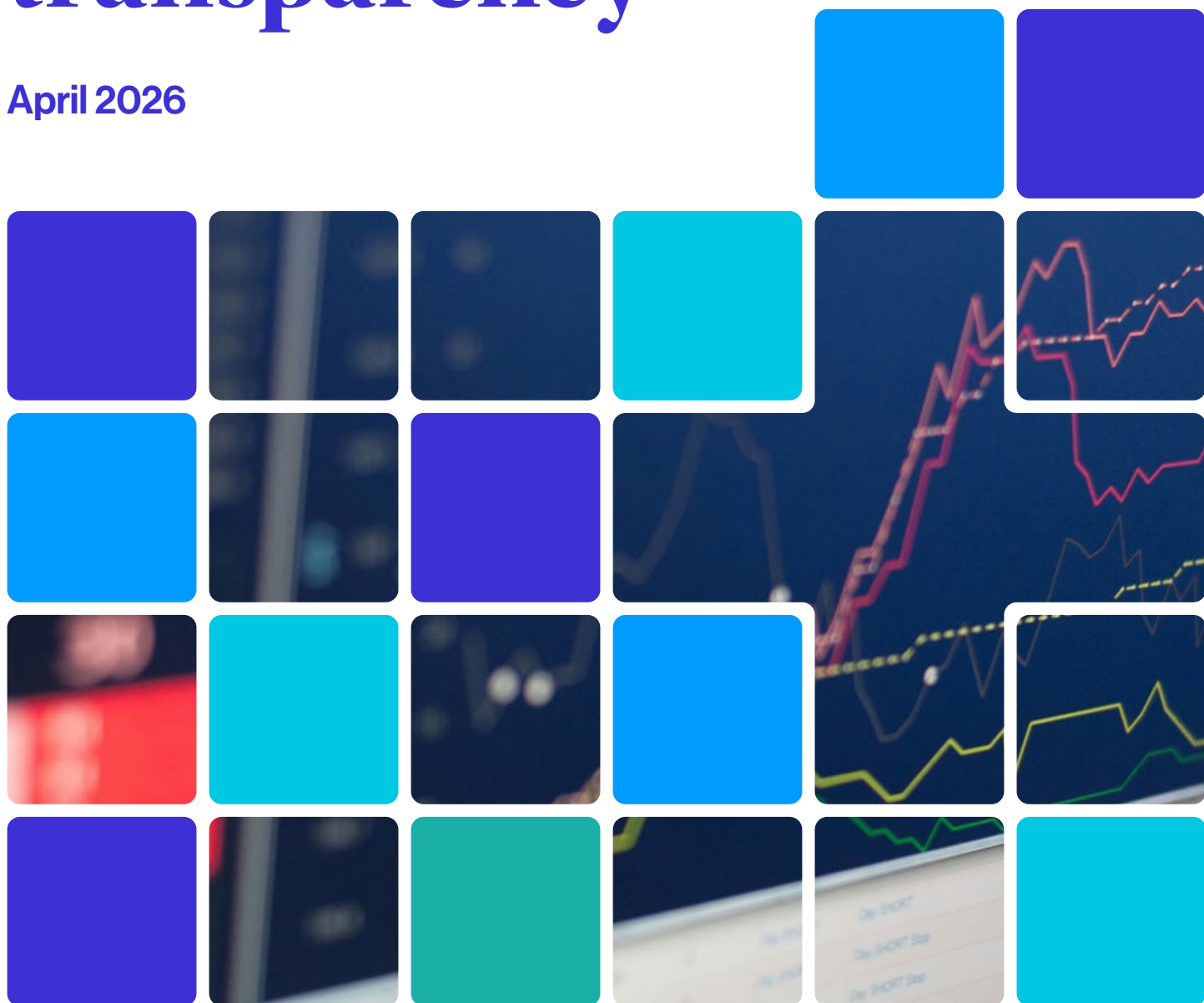
Working paper

Open
Ownership



Leveraging shareholder data for beneficial ownership transparency

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Contents

Summary	3
Introduction	4
Fragmentation in company ownership data systems	5
The case for better shareholder information	7
Compliance and accuracy	7
Understanding corporate networks	7
Improving the business environment	8
Analysis of shareholder and beneficial ownership information	9
What the analysis examines	9
What the data shows	11
Data overlap and alignment	11
Interest types in beneficial ownership data	14
Domesticity of corporate networks	16
Discussion and conclusion	17

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Summary

This working paper examines how and to what extent shareholder information can be leveraged to improve beneficial ownership transparency (BOT) and help understanding corporate structures or networks. While there is conceptual overlap between shareholding (i.e. legal ownership) and beneficial ownership (BO), they are often treated separately in legal and technical frameworks. This is despite the fact that high-quality, centralised shareholder data can strengthen BO disclosure regimes by **supporting verification; reducing compliance burdens and costs; and avoiding the collection of duplicate information**. It also **improves the usability** of BO data and helps users to **accurately understand complex corporate networks**. Better shareholder data also supports many of the policy objectives that governments pursue through BOT reforms.

Against this backdrop, this study assesses how shareholder register data can contribute to understanding corporate networks and inform the design and implementation of BOT systems. Drawing on two case studies – Estonia and Zambia – the analysis compares structured shareholder and BO data, and explores the following questions:

- **Data overlap and alignment:** To what extent do shareholder and BO registers capture the same ownership information?
- **Interest types in BO data:** What proportion of BO declarations relate to shareholding, compared to other forms of control (e.g. voting rights, appointment powers, or other means of control)?
- **Domesticity of corporate networks:** To what extent are corporate networks derived from shareholder data domestic, and how does this compare with those derived from BO data?

The analysis highlights distinct and complementary roles for shareholder and BO data in understanding corporate networks. Across both contexts, the findings reinforce that **most corporate structures are relatively simple**, often involving direct ownership by a small number of individuals. In such cases, **shareholder registers can already capture a substantial proportion of interests and relationships** that are later reported through BO declarations, suggesting that **shareholder data can play a significant role in supporting BOT systems** in practice.

At the same time, information collected through **BO declarations remains essential to capture forms of control that are not visible through shareholding**, including indirect ownership and other non-share-based mechanisms. In lieu of access to foreign ownership data, it helps identify cross-border ownership structures. Given the predominance of simple structures, these insights are likely applicable beyond Estonia and Zambia.

Overall, the findings suggest that leveraging existing – or investing in – central shareholder registers can significantly enhance the effectiveness of BOT reforms while also helping achieve those reforms' policy aims. While such improvements can be achieved at the national level for immediate benefits, greater alignment across jurisdictions and improved cross-border access to this information will strengthen understanding of transnational corporate networks and may help to gradually reduce the duplication in BO declarations at the national level.

Introduction

BO registers for legal vehicles play a critical role in BOT. Countries typically implement BOT reforms to support a range of objectives related to improved governance and accountability. Regardless of the specific objective, **understanding corporate structures or networks – namely, the relationship between individuals, legal vehicles, and assets – is essential for these reforms to succeed.** Collecting information about beneficial owners through declarations by legal vehicles is a key source of such data.

The effectiveness of these systems depends not only on the existence of reporting requirements but also on the quality, completeness, and usability of the data collected. Ensuring that BO information is accurate, consistent, and proportionate to risk remains a key challenge for policymakers and implementers.

Beneficial and legal ownership are often framed as distinct concepts, but they are tightly linked.¹ For example, an individual may be a beneficial owner based on their significant *legal* ownership of a legal vehicle, or they may be a beneficial owner based on other *interests* in it – direct or indirect – which enable them to control or benefit from its activities and holdings. Legal ownership is one of many types of interest which can lead to an individual being considered a beneficial owner of a legal entity. In reality, the vast majority of companies and similar legal vehicles are not part of wider corporate networks, vested in trusts, or subsidiaries of foreign companies. Most are owned and controlled directly by one or a few individuals, or have very simple corporate structures. In such cases, legal ownership is the main determinant of beneficial ownership.

Many jurisdictions have established BO registers for legal entities that operate separately from registers of their legal ownership, such as central shareholder registers. At the same time, legal entities in many jurisdictions are already required to report detailed shareholder information to company registers. Where this data is centralised, well structured, and regularly updated, it may capture a significant proportion of relationships subject to BO declarations, particularly in cases involving relatively simple ownership structures based on direct ownership of shares. Some jurisdictions, such as Switzerland, have proposed leveraging shareholder information in BO declarations.²

However, **how and the extent to which existing shareholder data can support, complement, or streamline BOT** has not yet been thoroughly examined.³ It is therefore important to explore the

¹ See, for example: Financial Action Task Force (FATF), *Guidance on Beneficial Ownership for Legal Persons* (FATF, 2023), 16, https://www.fatf-gafi.org/content/dam/fatf-gafi/guidance/Guidance-Beneficial-Ownership-Legal-Persons.pdf.co_redownload.pdf; Abu Dhabi Global Market (ADGM), “Beneficial ownership vs legal ownership” in *Guidance: Beneficial ownership and control* (ADGM, n.d.), <https://en.adgm.thomsonreuters.com/rulebook/beneficial-ownership-vs-legal-ownership>.

² Switzerland, State Secretariat for International Financial Affairs, “Verordnung über die Transparenz juristischer Personen und die Identifikation der wirtschaftlich berechtigten Personen”, 15 October 2025 to 30 January 2026, https://www.demokratis.ch/vernehmlassung/bund/verordnung-ueber-die-transparenz-juristischer-personen-und-die-identifikation-der-wirtschaftlich-berechtigten-personen_nn6bmp2d.

³ A 2013 report on the potential cost of BO declarations in the UK estimated that there were 225,000 unregistered beneficial owners not already covered by shareholder and director information. See: John Howell & Co. Ltd., *Costs of beneficial ownership declarations* (Global Witness, 2013), 7, <http://cdn-globalwitness-production.s3-website-eu-west-1.amazonaws.com/archive/files/library/cost%20of%20beneficial%20ownership%20declaration%20report.pdf>.

potential benefits of high-quality, centralised shareholder data for BOT; assess the degree of overlap between these information sources in terms of parties and interests; and estimate the prevalence of relatively simple ownership structures compared to complex ones. Addressing these questions will help inform the design of more effective BOT regimes.

Fragmentation in company ownership data systems

Information about legal entities in shareholder data and BO declarations show related aspects of corporate networks. Beneficial ownership refers to the natural person(s) who ultimately own, control, or benefit from a legal vehicle, including through indirect or non-share-based means. Shareholder data records the direct, legal owners of shares, who may be natural persons or legal vehicles. Generally, **the objectives of these registers may be different**. For example, shareholder registers may be under the general purview of a company registrar executing its mandate, while BO registers may be implemented for specific policy purposes, such as anti-money laundering (AML).

The owning of shares applies to companies as well as some other types of legal vehicles.⁴ A shareholder is an individual or legal entity that owns at least one share of a company's stock. Shareholders essentially own the company, and this comes with the right to share in the profits.⁵ While in some jurisdictions shareholder data is held by the companies themselves, in others this information may be centrally held by a company registrar.⁶ The time period within which shareholder information is or needs to be updated, and how the information is stored, its legal significance, differs across jurisdictions. For example, in the United Kingdom (UK), shareholder information is updated annually and is accessible via a PDF. In Zambia, information needs to be updated immediately upon share transfer, and it is held as structured data. In Pakistan, there is a proposal for the central shareholder register to become a digital central securities depository, where the registration of shares has a constitutive effect, giving legal certainty over ownership of the shares enabling their trading.⁷ However, different legal entities, such as non-profits, may sometimes be registered with different authorities.

A BO register is often overseen by the company registrar, but it may also be maintained by other authorities, such as tax authorities or financial intelligence units. Even where BO information is held by the company registrar, BO reporting obligations may extend to a wider range of legal entities than those covered by the company registrar. BO declaration systems collect varying amounts of information on intermediaries where declared interests are indirect (see Figure 1). This can range

⁴ For example, associations and partnerships in some jurisdictions. For simplicity, this paper will at times refer to "companies".

⁵ Adam Hayes, "Shareholder Definition: Rights & Types Explained", Investopedia, updated 2 April 2026, <https://www.investopedia.com/terms/s/shareholder.asp>.

⁶ For example, Australia, Austria, Belgium, China, Denmark, France, Germany, Greece, Ireland, Italy, Japan, Malaysia, Malta, the Netherlands, New Zealand, Singapore, and the UK.

⁷ For example, Pakistan: SECP, "SECP Approves Full Digitization of Share Ownership for Unlisted Companies to Enhance Transparency and Investor Protection", LinkedIn, March 2026, <https://www.linkedin.com/posts/secp-approves-full-digitization-of-share-share-743052263597402112-b1Zt>.

from not collecting any information on the indirect relationship, or just collecting the fact of whether the relationship is direct or indirect, to collecting information on some or all of the intermediaries.⁸

A beneficial owner is often defined in regulations as someone who meets one or more of a set of criteria. One such common criteria is *controlling ownership of a company through direct or indirect shareholding*. This is generally expressed in a percentage – often 25% – although there are many examples of jurisdictions adopting thresholds as low as 5%.⁹

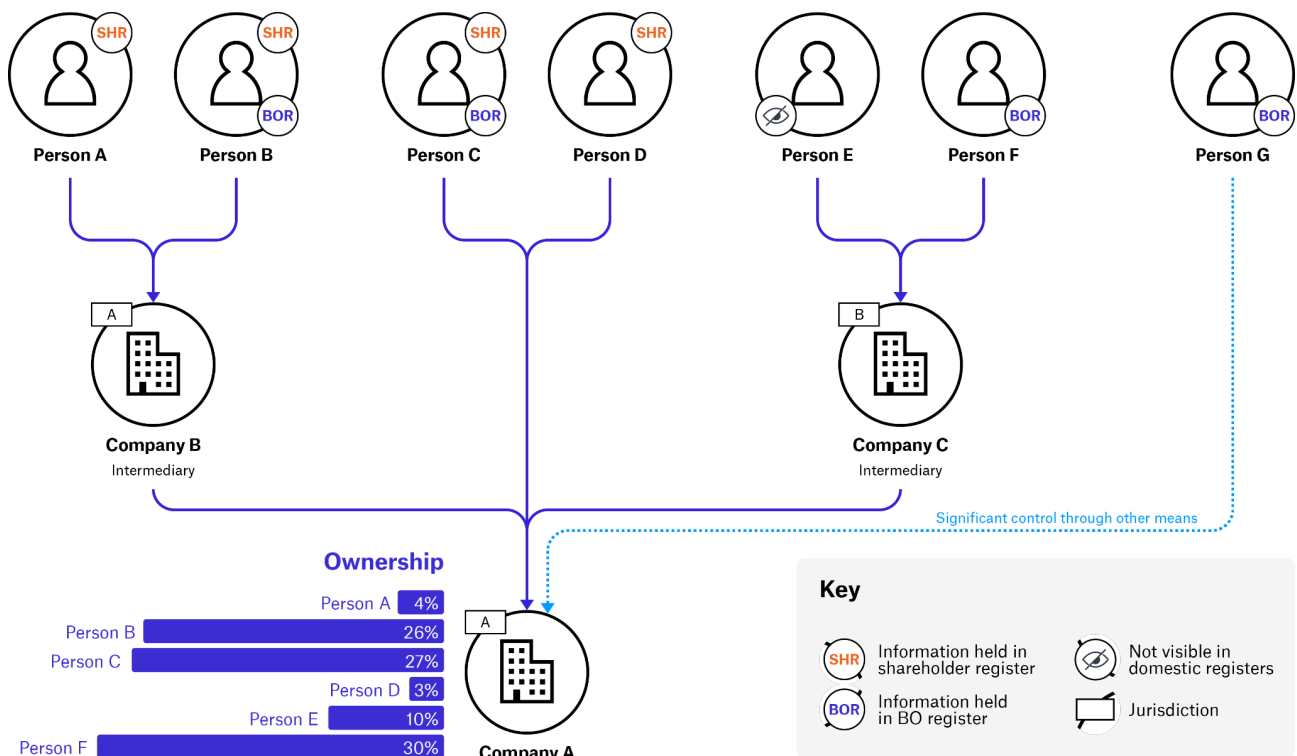
As a result, there are a number of corporate network types where beneficial owners can be determined solely on the basis of domestic records of legal ownership. Of these, the simplest is where a company's shares are all directly held by individuals. Those individuals whose shareholding exceeds the BO threshold are deemed beneficial owners. Another corporate network in this category is where individuals hold indirect shares above the BO threshold, but all companies in the network are domestic. BO information's added value is expected to be with more complex networks involving cross-border ownership (in the absence of international data-sharing), legal arrangements, and other forms of influence and control not based on the ownership of shares. These structures are of particular interest for tackling crime.

Shareholder data also provides additional granularity where share ownership is below the legal threshold for beneficial ownership, which means it is not included in BO information. By contrast, BO information will also include information about parties with non-share-based interests, or where ownership is held indirectly by shares through a foreign legal vehicle (see Figure 1).

⁸ For example, the UK does not state whether the relationship is held directly or indirectly. South Africa collects information on the parent of a declaring company in an indirect relationship. Armenia collects information about all intermediaries in an indirect relationship.

⁹ For example, Nigeria. The maximum threshold stipulated by the FATF is 25%. See: FATF, *International Standards on Combating Money Laundering and the Financing of Terrorism & Proliferation: The FATF Recommendations* (FATF, updated 2025), 96, <https://www.fatf-gafi.org/content/dam/fatf-gafi/recommendations/FATF%20Recommendations%202012.pdf.core.download.inline.pdf>.

Figure 1. Illustrative example of a corporate network revealed through domestic beneficial ownership and shareholder data (25% threshold)



In this corporate network, Company A's shares are directly owned by two companies: Company B (both registered in Jurisdiction A) and Company C (registered in foreign Jurisdiction B); and two individuals: Person C and Person D. Person A and Person B own shares in Company A indirectly via Company B, while Person E and Person F do so through Company C, making both Company B and Company C intermediaries in the network. Person G exercises significant control through other means over Company A. Depending on the nature of the interest (i.e. share ownership or other forms of control) and the level of ownership (i.e. percentage of shares owned), individuals and entities may appear in the domestic shareholder register, BO register, or both. Information can be combined from these sources to reveal this network. For example, Person F is identified in BO data because their ownership exceeds the 25% threshold, whereas Person E is not visible, as their shares are held through a foreign entity and fall below the threshold.

Despite this complementarity, **shareholder and BO registers are often treated as separate and disconnected data sources**. Jurisdictions implementing BO registers for legal vehicles may start from scratch, when in reality they may already be holding relevant information. This creates uncertainty around how the two datasets relate in practice, how much BO information is already observable through shareholder data, and where the two diverge.

These considerations are also critical for **facilitating cross-border data use and international data-sharing**, particularly in identifying transnational ownership structures that cannot be fully understood within a single jurisdiction and enabling ownership information from different sources to be meaningfully combined and analysed.

The case for better shareholder information

High-quality, centralised shareholder information can both **strengthen BO disclosure regimes** as well as have a **wide range of benefits** for the policy objectives BOT reforms are often seeking to achieve. This section provides a broad, top-level overview of these benefits.

Compliance and accuracy

Due to the overlap of shareholder and BO information, the former can be used to **verify BO declarations**, particularly for more complex, multilayered structures. BO declarations involving interests held through these types of structures are often unverifiable. Alternatively, shareholder information can be used to pre-populate or provide exemptions for BO declarations, thereby **lowering the compliance burden and cost for reporting companies** (particularly for those with a relatively simple ownership structure).¹⁰

In many of the policy objectives of reforms – e.g. tackling money laundering and illicit financial flows – BOT is often aiming to shed light on complex corporate structures, as these are where the risks are concentrated. While BOT targets complexity, in many jurisdictions there has been opposition to reforms due to the compliance burden on smaller businesses, and implementing agencies may seek ways to reduce these. In more complex structures, it will be easier for companies to report their direct owners, as they are more likely to accurately know who these parties are compared to the ultimate beneficial owners. This means information is also **more likely to be accurate and up to date**. Both of these use cases involve **consolidating duplicate information**, which may also benefit data users who may otherwise struggle with conflicting information.

Where jurisdictions hold companies' shareholder information centrally, and where these registers have a constitutive effect – i.e. being named in the register as owning a share confers legal certainty, in a similar way to asset registers – the information can be expected to be **highly accurate**.¹¹ These registers can serve as a single source of truth and a **backbone for digital public infrastructure**.¹²

Understanding corporate networks

Shareholder and BO data can also be combined to **understand full corporate networks**: structured representations of how companies, individuals, and other entities are linked through intermediaries

¹⁰ Niek Zaman and Hannah Erdman, *Een Centraal Aandeelhoudersregister (CAHR) als betrouwbaar en innovatief opstapje voor digitalisering van aandelen en een betere kapitaalinfrastructuur voor BV's* (Universiteit Leiden, 2020), 5–6, <https://www.universiteitleiden.nl/binaries/content/assets/rechtsgeleerdheid/instituut-voor-privaatrecht/cahr-onderzoek-finaal-rapport-dd-25-november-2020.pdf>.

¹¹ For BOT of assets, please see: Maria Jofre and Tymon Kiepe, *Bridging the gap for effective asset transparency: Analysing land registers and beneficial ownership data for legal vehicles* (Open Ownership, 2026), <https://www.openownership.org/en/publications/bridging-the-gap-for-effective-asset-transparency/>.

¹² Organisation for Economic Co-operation and Development (OECD), *Digital Public Infrastructure for Digital Governments*, OECD Public Governance Policy Papers No. 68 (OECD, 2024), 15, https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/12/digital-public-infrastructure-for-digital-governments_11fe17d9/ff525dc8-en.pdf.

(see, for example, Figure 1). This is particularly the case where shareholder information is accessible across borders.¹³ The availability of information on direct relationships is critical for building and combining these networks across data sources. Leveraging shareholder information may also help to provide **a level of detail not visible in BO declarations**, as well as **removing the need to lower BO reporting thresholds** to gain this detail.¹⁴ Accurately understanding corporate networks can be particularly valuable for financial crime investigations.¹⁵ This need also exists in use cases in public procurement, asset recovery, taxation, and customer due diligence (CDD).¹⁶

Improving the business environment

With respect to CDD, where company ownership information is highly accurate, particularly for simple structures, AML regulators can consider whether their **regulated entities can rely directly on this information** to meet their obligations for customer identification and verification. Rising compliance costs have made smaller loans – in the case of the Netherlands, up to EUR 100,000 – relatively costly for financial institutions.¹⁷ Changing CDD for simple, low-risk businesses may therefore **unlock access to loans and capital for small and medium-sized enterprises**.

These registers can also function as digital central securities depositories for the trading of shares. This can have benefits for **investor protection and helping investors make sound decisions**.¹⁸ It may make the buying and selling of shares in private limited companies easier and more efficient. While it may help provide access to capital through broader participation in private companies, it may also introduce systemic risks and ratchet up trade volumes. In addition, it could help with **organising quorate shareholder meetings**, a common challenge of smaller companies.¹⁹

Despite this range of benefits, there are no known jurisdictions that have implemented a comprehensive approach to company ownership information and understanding corporate networks.

¹³ For example, many commercial data service providers aggregate shareholder information to enable the understanding of corporate networks.

¹⁴ Tymon Kiepe, “Solving the international information puzzle of beneficial ownership transparency”, Open Ownership, 17 June 2024, <https://www.openownership.org/en/blog/solving-the-international-information-puzzle-of-beneficial-ownership-transparency/>.

¹⁵ Skatteetaten, *Konseptvalgutredning: Eierskapsopplysninger aksjer* (Skatteetaten, 2024), 20, <https://www.skatteetaten.no/contentassets/b480645c1aec4e6f89c1226a57a70e76/2024-12-kvu-eierskapsopplysninger-aksjer-.pdf>.

¹⁶ UK government, Procurement Act 2023, Schedule 6, Part 3, para. 45, <https://www.legislation.gov.uk/id/ukpga/2023/54/schedule/6/paragraph/45>; European Banking Authority (EBA), *Consultation Paper: Proposed Regulatory Technical Standards in the context of the EBA’s response to the European Commission’s Call for advice on new AMLA mandates* (EBA, 2025), <https://www.eba.europa.eu/sites/default/files/2025-03/gbc83e61-e9a1-4e91-93de-2af8325e0182/Consultation%20Paper%20on%20Response%20to%20Call%20for%20Advice%20on%20new%20AMLA%20mandates.pdf>; Isabelle Büchner, *Chasing Grand Corruption* (Transparency International, 2025), <https://files.transparencycdn.org/images/Report-Chasing-Grand-Corruption-English-2025-Oct.pdf>; Tymon Kiepe, *Leveraging information about ownership networks to improve taxation* (Open Ownership, 2025), <https://www.openownership.org/en/publications/leveraging-information-about-ownership-networks-to-improve-taxation/>.

¹⁷ Zaman and Erdman, *Een Centraal Aandeelhoudersregister*, 4.

¹⁸ SECP, “SECP Approves Full Digitization of Share Ownership”; Skatteetaten, *Konseptvalgutredning*, 31.

¹⁹ Zaman and Erdman, *Een Centraal Aandeelhoudersregister*, 4–5.

Analysis of shareholder and beneficial ownership information

This section assesses how and to what extent shareholder data can contribute to understanding corporate networks and inform the design and implementation of BOT systems.

What the analysis examines

The analysis is guided by three questions:

1. **Data overlap and alignment:** To what extent do shareholder and BO registers capture the same ownership information?

This analysis compares the number of legal entities and individuals recorded in each dataset; identifies overlaps between the two sources; and assesses what information is shared versus uniquely captured. It helps evaluate the completeness, redundancy, and structural alignment of the two systems, as well as the limits of shareholder data in capturing beneficial ownership.

2. **Interest types in BO data:** What proportion of BO declarations relate to shareholding, compared to other forms of control (e.g. voting rights, appointment powers, or other means of control)?

This question assesses the relative importance of share-based ownership versus non-share-based control; the structural limits of shareholder data as a proxy for beneficial ownership; and the types of interests which are either captured only in BO data or only in shareholder data.

3. **Domesticity of corporate networks:** To what extent are corporate networks derived from shareholder data domestic, and how does this compare with those derived from BO data?

This analysis examines the geographical scope of corporate networks in each dataset, assessing their relative capacity to identify foreign ownership and cross-jurisdictional chains. It highlights where shareholder data may be structurally limited in capturing transnational ownership.

Case studies

This analysis relies on access to structured BO and shareholder data. Therefore, the study is limited to two case studies: Estonia and Zambia.

Estonia

In **Estonia**, the electronic e-Business Register (*e-Äriregister*), managed by the Centre of Registers and Information Systems (RIK) includes a centralised shareholder register. With some exceptions, a company's list of shareholders is generally kept by the e-Business Register, meaning shareholders' data has legal meaning, and is therefore deemed to be up to date and accurate. To illustrate, shares are

deemed to have been issued and transferred when an entry is made in the e-Business Register.²⁰ Some legal entity types are registered elsewhere, such as with the register of non-profit associations and foundations, but are still required to declare beneficial ownership to the e-Business Register.²¹ By contrast, public limited companies are registered with the e-Business Register but register their shares in the Estonian Central Register of Securities.²²

The Estonian BO register for legal entities is an integral part of the electronic e-Business Register. The obligation has been effective since 2018, leveraging Estonia's **advanced e-governance infrastructure**. All legal entities registered in Estonia, with few exceptions, are required to identify, record, and disclose up-to-date information about their beneficial owners within 30 days of a change. This is governed by the Money Laundering and Terrorist Financing Prevention Act.²³

Both datasets are publicly available for download via online portals, bulk downloads, and API services.

²⁴ For this paper, the datasets were downloaded on 11 February 2026 and matched using a hashed personal identifier for individuals and a company registry code for legal entities, enabling precise comparison between shareholder and BO records. All interests are assumed to be current (the end date is null for all entries).

Zambia

In **Zambia**, both the shareholder register and BO register for legal entities are managed by the Patents and Companies Registration Agency (PACRA). Shareholdings are recorded by companies themselves and considered issued when the company updates its own register. Companies have an obligation to update PACRA of any changes within 14 days, and are assumed to be relatively accurate and up to date.

Zambia's BO disclosure regime is undergoing changes in response to a number of shortcomings in the legislation and systems. While Zambia amended its Companies Act in 2025, the current BO dataset was collected under the previous legislation.²⁵ Although beneficial ownership is broadly defined within the previous Act, it is mostly mentioned in respect to shares or shareholding.²⁶ In the system, declarants are asked to disclose the shareholders of a company, then to declare beneficial ownership of those shares. The legislation has been amended and Zambia also plans to capture beneficial ownership with non-shareholder interests. Companies similarly need to disclose updated BO information to PACRA within 14 days of a change.

²⁰ Riigi Teataja, Commercial Code, 27 February 2024, <https://www.riigiteataja.ee/en/eli/ee/511012016002/consolide/current>.

²¹ European e-Justice Portal, "Business registers in EU countries: Estonia", updated 20 September 2022, https://e-justice.europa.eu/topics/registers-business-insolvency-land/business-registers-eu-countries/ee_en.

²² EESTI.EE, Riigiportaal, "Founding a foundation", updated 13 March 2024, <https://www.cesti.ee/erisik/en/artikkel/doing-business/establishing-a-company/founding-a-foundation>; Invest in Estonia, "Public limited company (AS)", n.d., <https://investinestonia.com/business-in-estonia/establishing-company/legal-forms/limited-public/>.

²³ Riigi Teataja, "Money Laundering and Terrorist Financing Prevention Act", 17 November 2017, <https://www.riigiteataja.ee/en/eli/517112017003/consolide>.

²⁴ Shareholders: E-Business Register, home page, n.d., <https://ariregister.rik.ee/eng>. Beneficial owners: E-Business Register, "Beneficial owners", n.d., https://ariregister.rik.ee/eng/beneficial_owners_query. Also see: E-Business Register Open Data, "Shareholders", n.d., <https://avaandmed.ariregister.rik.ee/en/downloading-open-data#pk-6>.

²⁵ Mulenga Mundashi Legal Practitioners, "Zambia Introduces Sweeping Corporate Transparency Reforms Targeting Beneficial Ownership", Afriwise, 21 January 2026, <https://www.afriwise.com/blog/zambia-introduces-sweeping-corporate-transparency-reforms-targeting-beneficial-ownership>.

²⁶ Zambia Legal Information Institute, Companies Act, 2017, 20 November 2017, <https://zambialii.org/akn/zm/act/2017/10/eng@2017-11-20#page-1>.

Recent evidence indicates a relatively low compliance rate, with an estimated 38–44% of companies having submitted BO information.²⁷ PACRA is undertaking **reforms to modernise the company and BO registers** in partnership with Open Ownership to improve data quality as well as strengthen compliance and enforcement mechanisms.²⁸ Through this partnership, anonymised shareholder and BO datasets were provided by PACRA on 13 February 2026, with hashed personal identifiers and hashed company numbers. As Zambian data does not include an end date field, all interests are assumed to be current.

Limitations

Together, these cases provide a complementary basis for examining how shareholder and BO data interact across different institutional and data contexts. The study's main limitation is the small sample, given the challenges in accessing both structured shareholder and BO data for a jurisdiction. Additionally, there are limitations due to the characteristics of the Zambian BOT regime discussed above.

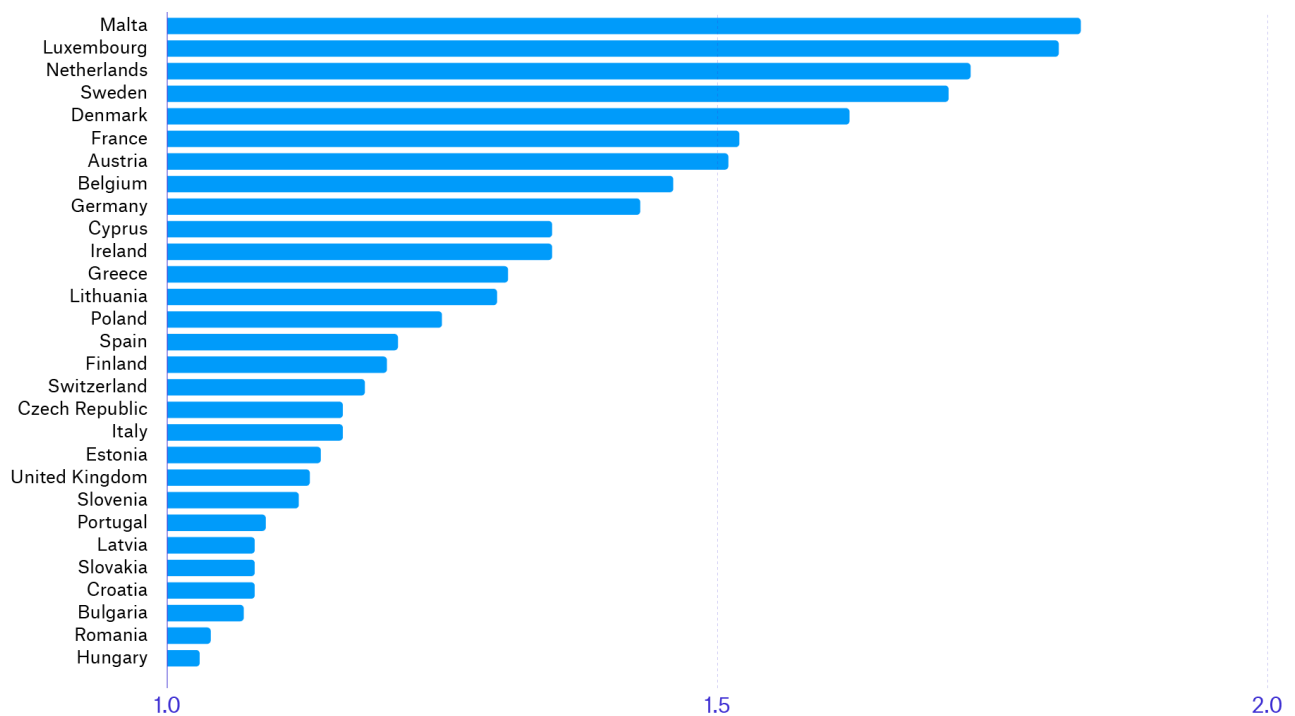
Despite these limitations, there are reasons to believe these insights are more broadly applicable. Ownership structures appear to be relatively simple for most companies, with beneficial owners holding direct interests in a company rather than through multiple layers of intermediary entities. Across Europe, for instance, beneficial owners typically hold direct stakes – the corporate structures have one layer – suggesting that most of these relationships are already visible through shareholder information (see Figure 2).²⁹

²⁷ Augustine Sichula, “PACRA unveils three-year plan to modernize Zambia’s beneficial ownership registers, promote integrity”, *Zambia Monitor*, 23 January 2026, <https://www.zambiamonitor.com/pacra-unveils-three-year-plan-to-modernize-zambias-beneficial-ownership-registers-promote-integrity/>; Lazarous Tembo, “PACRA, Partners Roll Out Major Business Ownership Transparency Project”, *Efficacy News*, 26 January 2026, <https://efficacynews.africa/2026/01/26/pacra-partners-roll-out-major-business-ownership-transparency-project/>. While compliance rates are increasing and these rates may not fully correspond to the date the dataset was retrieved, it is unlikely to have increased significantly.

²⁸ Sichula, “PACRA unveils three-year plan”; Miranda Evans, “Beneficial ownership and business register reform in Zambia: Announcing our new partnership with PACRA”, Open Ownership, 6 March 2026, <https://www.openownership.org/en/blog/beneficial-ownership-and-business-register-reform-in-zambia-announcing-our-new-partnership-with-pacra/>.

²⁹ Antonio Bosisio, Carlotta Carbone, Maria Jofre, Michele Riccardi, and Stefano Guastamacchia, *Project DATACROS: Developing a Tool to Assess Corruption Risk factors in firms’ Ownership Structures* (Transcrime – Università Cattolica del Sacro Cuore, 2021), 29–30, https://www.transcrime.it/wp-content/uploads/2021/09/Datacros_report.pdf.

Figure 2. Average number of layers between a reporting company and its beneficial owners in Europe (where 1 = direct ownership)³⁰



What the data shows

Data overlap and alignment

Table 1 compares the shareholder (SH) and BO datasets in the Estonia and Zambia case studies, showing the number of companies and individuals recorded in each dataset, as well as the extent of overlap and dataset-specific records. The figures cover companies and individuals only, and exclude corporate shareholders. The results reveal **markedly different patterns across the two contexts**.

Table 1. Coverage and overlap between shareholder and beneficial ownership datasets (Estonia and Zambia)

Country	Unit of analysis	SH dataset	BO dataset	Overlap (BO/SH)	Only in SH	Only in BO
Estonia	Companies (registry code)	280,862	295,010	266,868	13,994	28,142
	Individuals (hashed ID)	237,347	246,008	211,516	25,831	34,492
Zambia	Companies (hashed company number)	555,809	104,689	104,637	451,172	52
	Individuals (hashed ID)	699,821	193,212	189,940	509,881	3,272

³⁰ Adapted from: Bosisio et al., *Project DATAACROS*, 29.

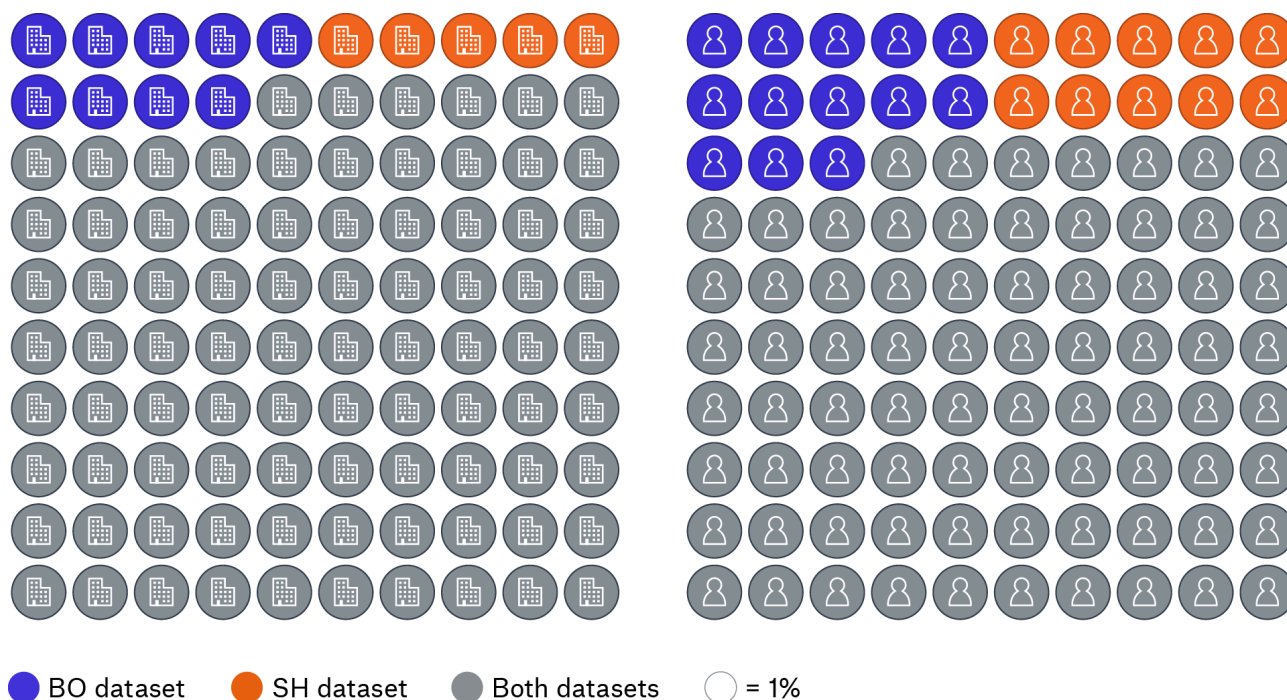
Estonia

Estonia shows **substantial overlap between shareholder and BO datasets**, with both sources covering largely similar populations of companies and individuals (Figure 3). Most records appear in both datasets, indicating strong structural alignment between the two systems.

At the same time, each dataset contains a subset of companies and individuals not present in the other. This indicates that, while shareholder data captures a significant share of the ownership relationships later reported through BO declarations, **BO data provides additional, distinct information** not observable through shareholder records alone.

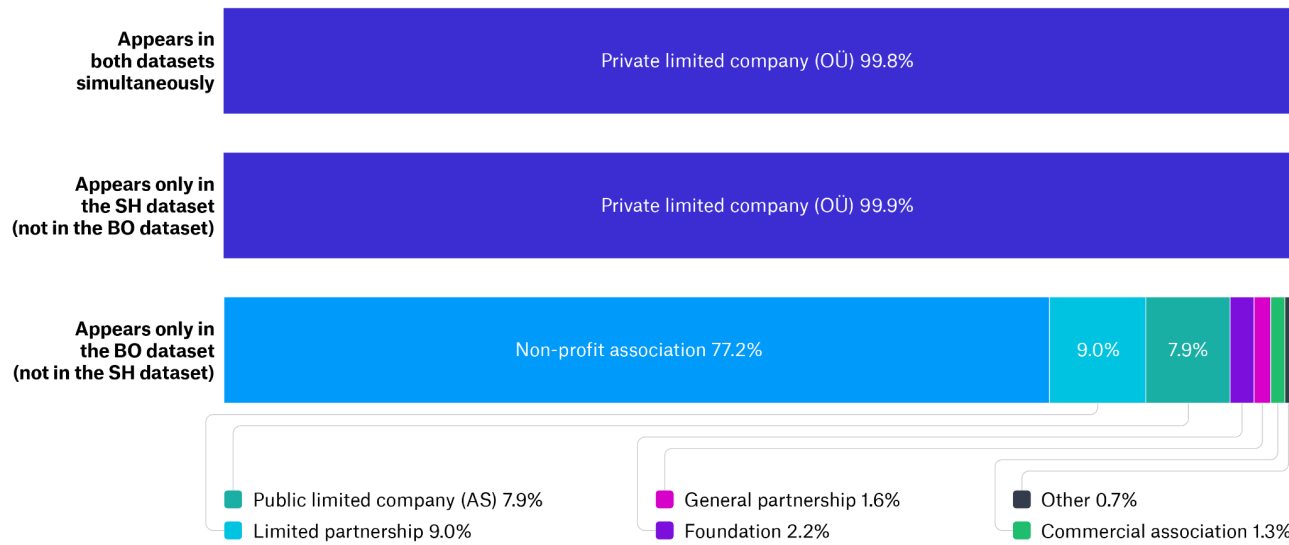
Of the 25,831 individuals only present in the Estonian shareholder dataset, 39% (almost 10,000 individuals) hold ownership shares of 25% or less (the threshold for being considered a beneficial owner). The remaining 61% (15,849 individuals, or approximately 7% of all individuals in the shareholder data) report shares above this threshold. Further research may be required to explain this latter group. Possible reasons include noncompliance with BO reporting, differences in reporting timelines, exemptions to BO reporting requirements, or other factors not yet explored.

Figure 3. Distribution of companies and individuals across datasets (Estonia)



Entities appearing in both datasets, as well as those only captured in shareholder data, are predominantly private limited companies. By contrast, **BO data covers a wider range of legal entity types** (Figure 4). These entities may not have shareholders (such as foundations) or may register their shares in separate systems. As a result, shareholder data primarily captures ownership information for private limited companies, although this is the most common entity type.

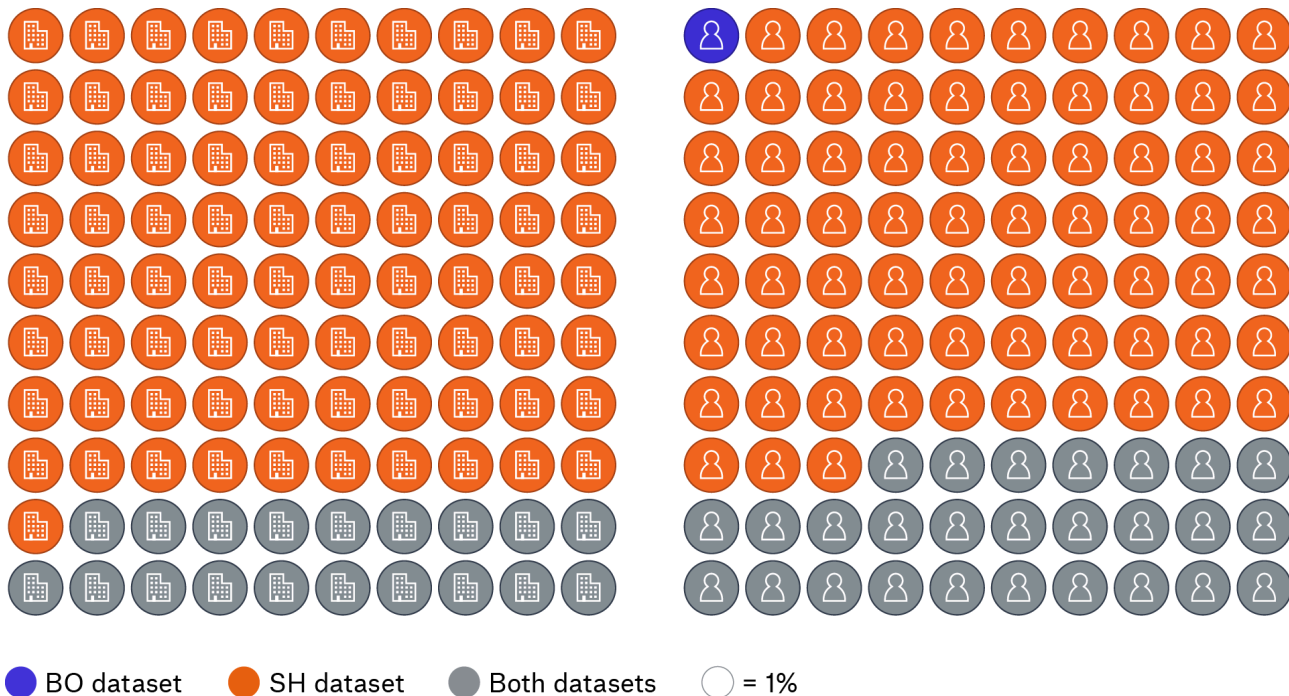
Figure 4. Distribution of company legal forms across datasets (Estonia)



Zambia

The Zambian case presents a markedly different configuration (Figure 5). Nearly all companies and individuals recorded in the BO register can be matched to shareholder records. By contrast, the shareholder dataset includes a very large number of additional entities and individuals not reflected in BO declarations. As a result, BO records represent a relatively small subset of the overall population captured in shareholder data.

Figure 5. Distribution of companies and individuals across datasets (Zambia)



This pattern suggests that many companies recorded in the shareholder register are not captured in the BO dataset, indicating that **BO information is missing for a substantial part of the registered population**. While part of this difference may reflect reporting thresholds or other factors, the scale of the discrepancy is also consistent with incomplete compliance with BO disclosure requirements.

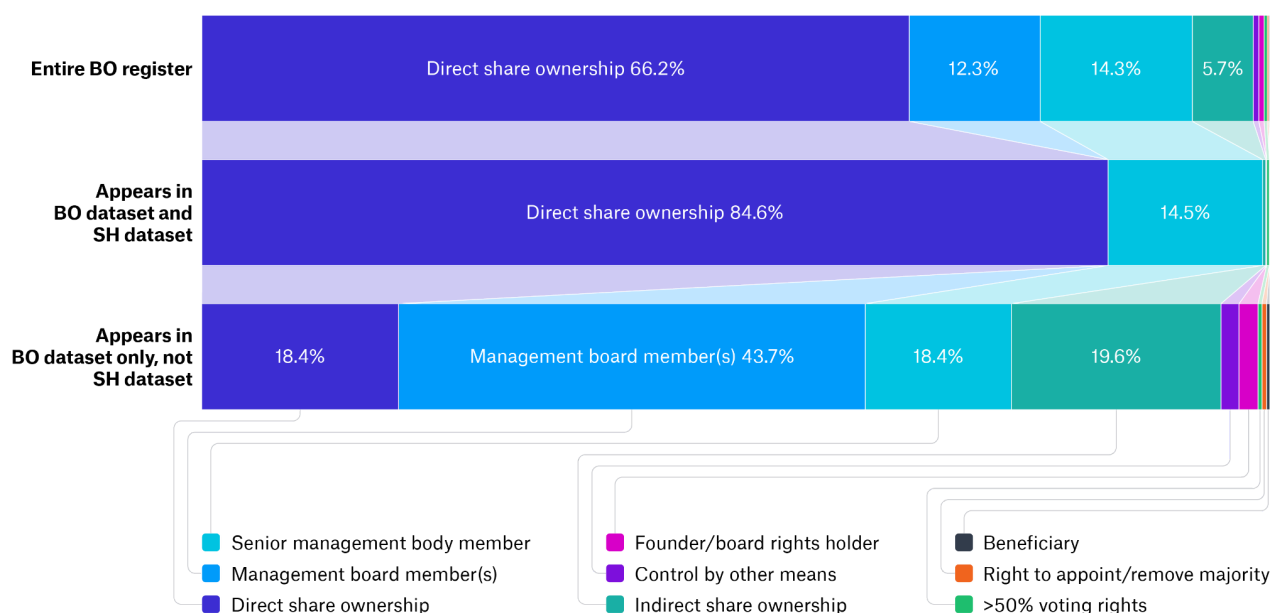
Interest types in beneficial ownership data

Corporate networks are often relatively simple and based on direct shareholding by a small number of individuals. In such cases, a high degree of overlap between shareholder and BO data is expected. However, BO data also captures forms of control that are not reflected in shareholding, including indirect ownership and other means of control.

Estonia

In Estonia, BO data includes several types of declared interests, with direct share ownership representing the largest category (Figure 6). Amongst the companies that are also represented in the shareholder register, direct share ownership and senior management roles are proportionally more prominent. By contrast, **companies only appearing in the BO dataset show a higher relative share of non-share-based interests**, particularly management board membership and indirect share ownership. Instances of direct share ownership appearing only in the BO data may reflect entity types whose shares are not recorded in the business register.

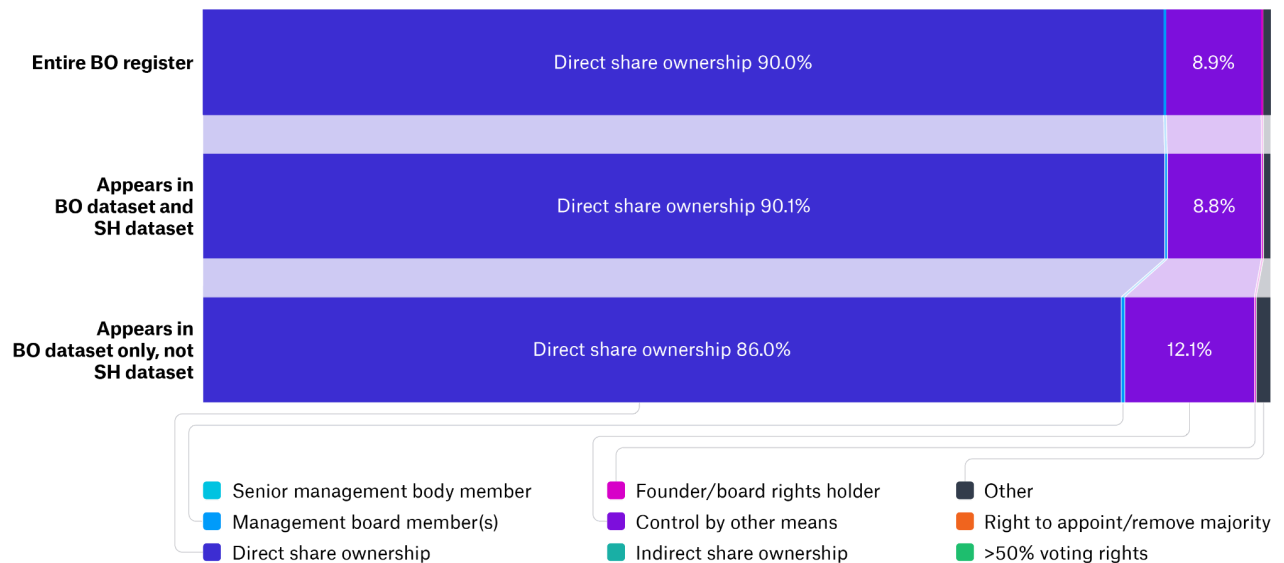
Figure 6. Distribution of interest types in Estonia's beneficial ownership dataset



Zambia

In Zambia, the distribution is more heavily concentrated on direct ownership, consistent with the broader coverage of shareholder data observed earlier (Figure 7). However, **within the BO-only segment, there is a relatively greater presence of other forms of control**, including control by other means, management board membership, and voting or appointment rights.

Figure 7. Distribution of interest types in Zambia's beneficial ownership dataset



These results reflect the deficiencies in the former Zambian BO disclosure regime outlined above, with information about BO being collected in relation to shareholdings. Additionally, the dataset records interest types in free-text fields, resulting in inconsistent categories, variations in terminology, and typographical errors. While the data was processed and mapped into broader categories based on the available descriptions, this introduces uncertainty in the classification. This highlights the importance of structured data collection using standardised categories, aligned with legal definitions, to enable consistent analysis and comparison across datasets.

Domesticity of corporate networks

While many corporate networks are relatively simple, they can also involve multiple layers of intermediary entities and links across jurisdictions. These cross-border elements may arise through foreign intermediary companies or through individuals who are non-residents or hold foreign nationality.

For the purposes of this analysis, a corporate network is defined as the set of ownership relationships linking a given legal entity to its direct and indirect beneficial owners. Each legal entity is treated as the starting point of a distinct network, even where ownership chains overlap. For instance, where one legal entity is owned by another legal entity, which in turn is owned by an individual, this is treated as two separate networks: one centred on the first entity (with multiple ownership layers) and one centred on the intermediary entity (with a single ownership layer). This approach allows for consistent comparison across entities and avoids combining partially overlapping structures.

In this context, domestic networks refer to ownership structures in which all relevant parties – both companies and individuals – are located within the same jurisdiction. Distinguishing between domestic and cross-border networks is key to understanding the geographical reach of ownership structures and the extent to which they can be fully observed within a single country.

Table 2 presents comparative statistics on the domesticity and cross-border characteristics of corporate networks based on shareholder and BO data in Estonia and Zambia. As above, the results reveal contrasting patterns across the two contexts.

Table 2. Domestic and cross-border characteristics of corporate networks (Estonia and Zambia)

Statistic	Estonia (n = 309,004)		Zambia (n = 555,861)	
	Count	%	Count	%
Entities in domestic networks	270,654	88%	476,189	86%
Entities where all beneficial owners are domestic	260,353	84%	78,072	14%
Entities where all individual shareholders are domestic	266,812	86%	476,372	86%
Entities where all corporate shareholders are domestic	23,934	8%	13,327	2%
Entities only present in SH data that only have domestic individuals as shareholders	7,384	2%	396,382	71%
Entities with foreign beneficial owners	35,406	11%	29,148	5%
Entities with foreign shareholders (total)	11,497	4%	79,136	14%
└ of which: foreign corporate shareholders	4,751	2%	4,404	1%
└ of which: foreign individual shareholders	7,384	2%	78,795	14%
Entities with foreign beneficial owners and only domestic shareholders	26,853	9%	536	0%

Estonia

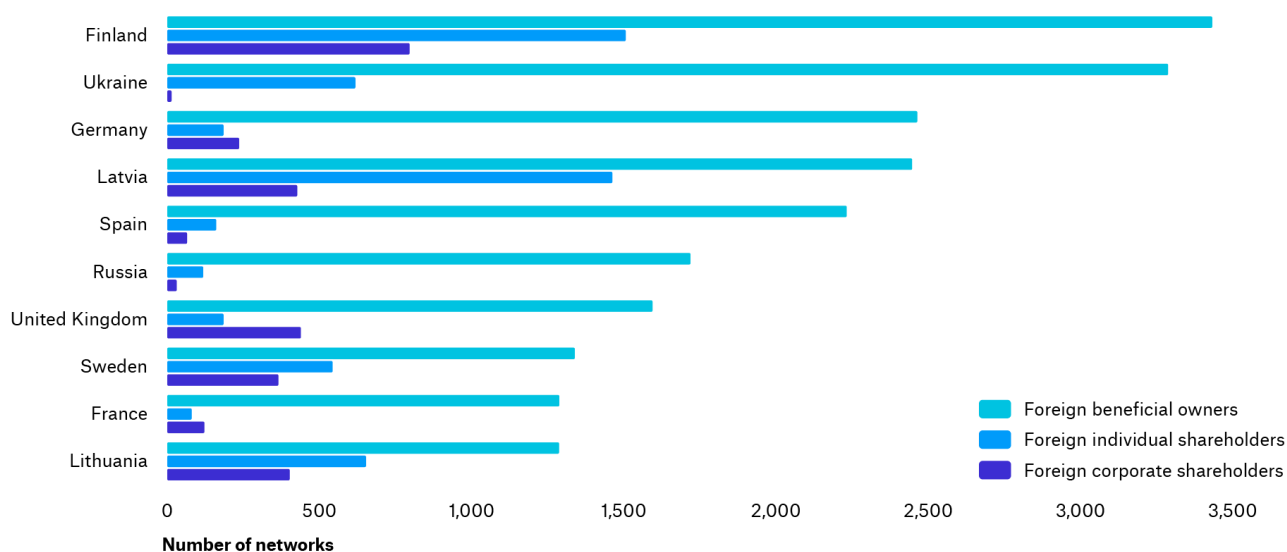
In Estonia, the vast majority of corporate networks are domestic, with 88% of entities showing no links to foreign jurisdictions through either intermediary entities or individuals. This aligns with the fact that 98% of entities do not have foreign shareholders.

At the same time, 4% of companies have foreign shareholders (either corporate or individual), while 11% have foreign beneficial owners. Notably, 9% of networks consist entirely of domestic shareholders (companies or individuals) but include at least one non-domestic beneficial owner, suggesting that **cross-border elements may only become visible at the BO level** – except where foreign individuals hold shares directly, in which case the full network remains visible within domestic shareholder data.

Only 8% of corporate structures involve domestic corporate shareholders, while 2% involve foreign corporate shareholders. This means the majority of **multi-layered networks are domestic**, and their intermediaries can be understood through domestic shareholder data.

Overall, shareholder data presents a more domestic view of ownership structures, while BO data reveals additional cross-border links not observable through shareholding alone, albeit for a relatively small proportion of networks (Figure 8). This pattern is particularly pronounced for connections with France, Germany, Russia, Spain, and Ukraine, where foreign links are more frequently captured in BO data. By contrast, countries such as Finland, Latvia, Lithuania, and Sweden show a more balanced distribution across both datasets.

Figure 8. Top 10 foreign countries in corporate networks (Estonia)



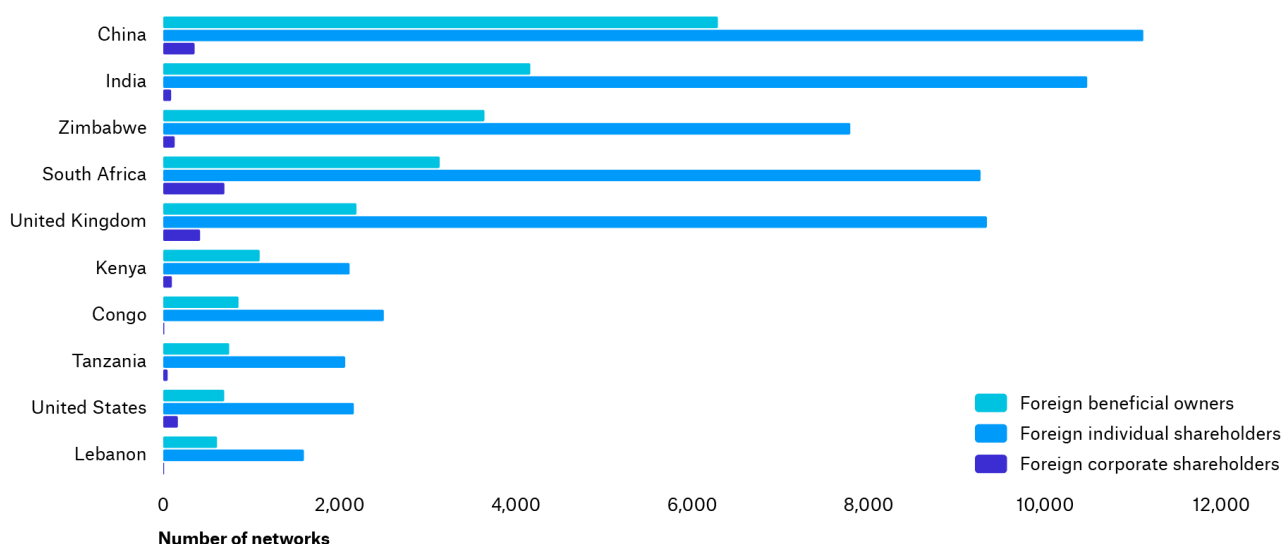
Zambia

In Zambia, most corporate networks are also domestic, with 86% showing no links to foreign jurisdictions. While lower BO reporting compliance may overestimate this number, 99% of entities do not include foreign companies as shareholders. However, the small proportion of complex, cross-border networks are disproportionately relevant to policy objectives such as fighting financial crime, underscoring the need for effective data-sharing across jurisdictions.

A substantially higher number of entities include foreign shareholders (79,136) – either corporate or individual – than foreign beneficial owners (29,148). This likely reflects gaps in BO reporting rather than an absence of cross-border networks. In this context, shareholder data may provide a more reliable starting point for identifying cross-border links and supporting compliance with BO reporting. Because 71% of entities only present in shareholder data just have domestic individuals as shareholders, PACRA may already have BO information for almost 400,000 companies that have not reported their beneficial owners. Although not all of these companies may be subject to BO reporting requirements, and some may be inactive, it indicates that leveraging shareholder information also has the potential to contribute significantly to current compliance levels.

BO data in Zambia presents a more domestic view of ownership structures, while **shareholder data reveals additional cross-border links** (Figure 9). Cross-border links in BO data are more concentrated in a smaller set of jurisdictions, particularly China, India, South Africa, the UK, and Zimbabwe.

Figure 9. Top 10 foreign countries in corporate networks (Zambia)



Discussion and conclusion

The analysis has helped answer the following research questions:

1. Data overlap and alignment: To what extent do shareholder and BO registers capture the same ownership information?

Across both case studies, there is **significant overlap between shareholder and BO registers**. The findings reinforce that many corporate structures are relatively simple, often involving direct ownership by a small number of individuals. In such cases, shareholder registers can already capture a substantial share of interests and relationships that are later reported through BO declarations. This suggests that shareholder data can play a significant role in supporting BOT systems, including by **improving accuracy and usability, reducing duplicate reporting, and lowering reporting burdens** by leveraging existing information.

2. Interest types in BO data: What proportion of BO declarations relate to shareholding, compared to other forms of control (e.g. voting rights, appointment powers, or other means of control)?

While the vast majority of interests relate to shareholding, information collected through BO declarations remains **essential to capture forms of control not visible through shareholding**, including indirect cross-border ownership and other non-share-based mechanisms.

3. Domesticity of corporate networks: To what extent are corporate networks derived from shareholder data domestic, and how does this compare with those derived from BO data?

Corporate networks mostly comprise direct ownership by natural person shareholders, meaning **domestic shareholder data can play a significant role in increasing compliance** – such as in Zambia – or the **verification of BO declarations**. In lieu of access to foreign shareholder data, BO data helps identify cross-border ownership structures. While complex structures may be a small proportion of corporate structures, these are the ones of primary interest in many policy use cases for BOT, including fighting financial crime. The analysis further suggests that, even when shareholder data from multiple jurisdictions is combined, it may still underestimate cross-border

links, as BO data captures additional relationships. This is especially the case for those involving foreign individuals and interests beyond the ownership of shares.

This work highlights the distinct and complementary roles for shareholder and BO data, and shows that shareholder data could be beneficial for improving BO information and better achieving policy goals. These findings suggest that agencies in charge of companies and BOT reforms should explore:

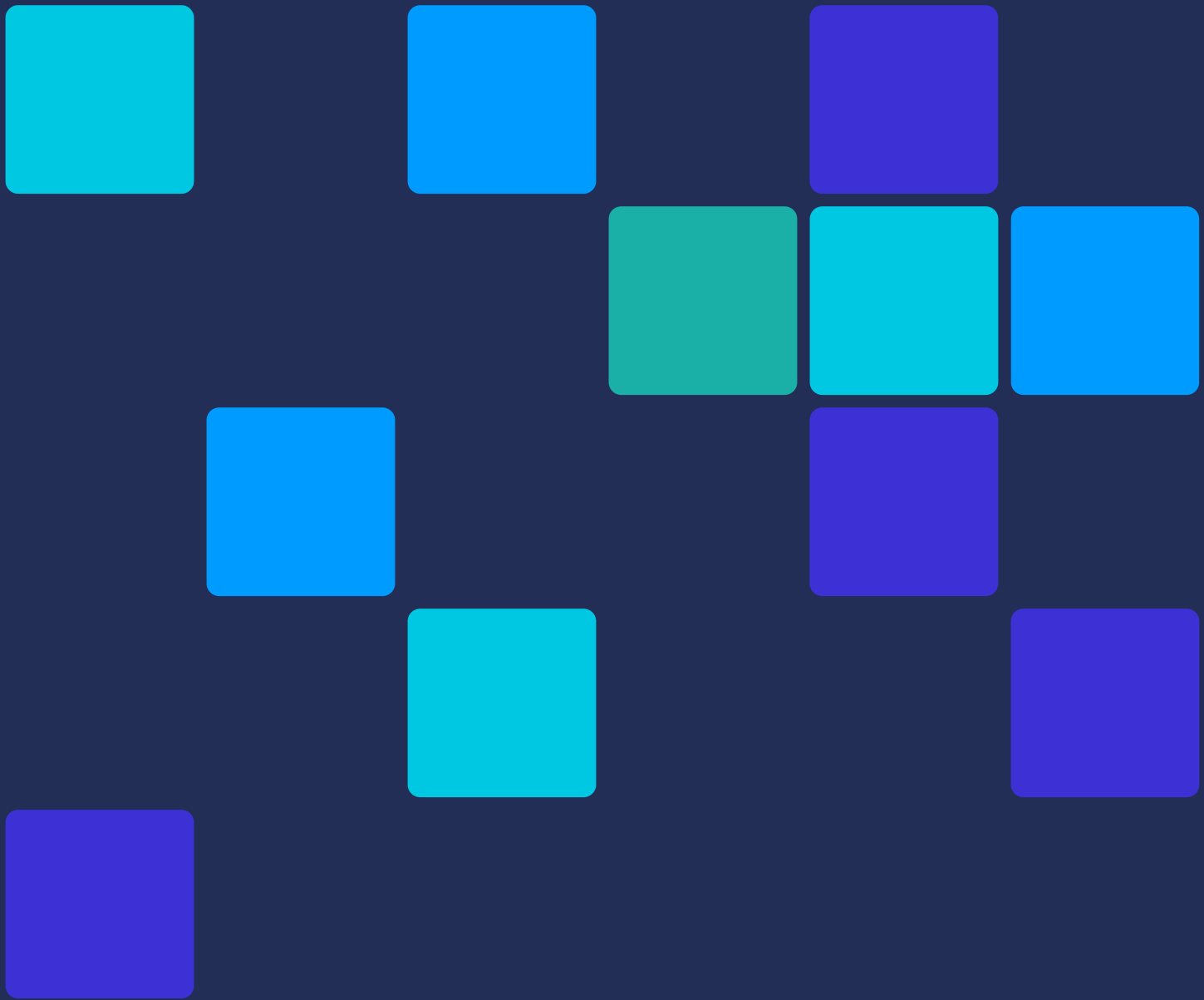
- Where possible, **conducting similar analyses** for their jurisdictions as those laid out in this paper, in order to help quantify the **potential value of shareholder information** and identify next steps, or consider making **anonymised datasets available** for research.
- Investing in existing or creating new **central shareholder registers** with up-to-date, accurate, and structured information.
- **Aligning shareholder and BO regimes**, including coverage of legal entities and reporting periods.
- **Reducing the duplication** of data collection through pre-populated forms, verification, or exemptions, particularly for simple corporate structures.
- In addition to efforts to **strengthen cross-border access** to BO information, improving this for shareholder information in order to help understand transnational corporate structures in more detail.
- Potentially, in future and as data-sharing improves, gradually **reducing data duplication at the international level**, e.g. by expanding exemptions to BO declarations for cross-border corporate structures covering jurisdictions where information is readily available.

Further research

Although Estonia and Zambia represent different contexts, further research is needed to see whether patterns of corporate networks are consistent across more jurisdictions, and whether the findings in this paper are indeed broadly applicable.

The analysis has shown a lack of alignment between the policy and legislative frameworks governing the collection and storage of shareholder and BO information. It was not always possible to explain certain differences in the data, for example: why a significant number of individuals appeared only in shareholder data that was not explainable due to a lack of compliance or below-threshold ownership shares. Further comparative legal and policy analysis will help shed light on these questions.

Finally, further research should examine jurisdictions currently exploring or implementing digital central shareholder registers, such as Norway, and with constitutive effect, such as Pakistan. It is also worth learning from jurisdictions currently implementing reforms that leverage shareholder data for BOT reforms, including Switzerland and Zambia.



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